assign-day-10

July 18, 2023

1 Assignment Problems: Day 10

The following problems require access to the file named 'Dataset_Day10.csv', provided with this assignment. (Use Random State = 50)

This is the same assignment data as Day 9.

The datasets include data from 768 women with several medical predictor variables and one target variable('Outcome'). The classification goal is to predict whether or not the patients in the dataset have diabetes or not.

For columns: Glucose, BloodPressure, BMI, DiabetesPedigreeFunction

If the column value is 0, then they should be considered as missing data.

[1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn import svm
from sklearn import tree
from math import log, sqrt
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve, log_loss, precision_recall_curve,␣
 ↪accuracy_score, precision_score, recall_score, f1_score
from sklearn.preprocessing import StandardScaler
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix
```

- Firstly, replace all Missing values with relevant figures.
- Then remove all existing outliers and get the final data for classification.

[2]:
```
data = pd.read_csv("Dataset_Day7.csv")

# Missing data treatment
print('Before Missing Data Treatment')
for col in ['Glucose', 'BloodPressure', 'BMI', 'DiabetesPedigreeFunction'] :
    print(col,' = ',len(data[col][data[col]==0]))
for col in ['Glucose', 'BloodPressure', 'BMI', 'DiabetesPedigreeFunction'] :
    data[col][data[col]==0] = data[col].median()
```

```
print('\nAfter Missing Data Treatment')
for col in ['Glucose', 'BloodPressure', 'BMI', 'DiabetesPedigreeFunction'] :
    print(col,' = ',len(data[col][data[col]==0]))

outlier = pd.DataFrame(columns =␣
 ↪["Pregnancies","Glucose","BloodPressure","BMI","Age"])
temp = data.drop('Outcome',axis=1)
for col in ["Pregnancies","Glucose","BloodPressure","BMI","Age"]:
    Q1 = temp[col].quantile(0.25)
    Q3 = temp[col].quantile(0.75)
    IQR = Q3 - Q1
    UpperBound = Q3 + 1.5*IQR
    LowerBound = Q1 - 1.5*IQR
    outlier[col] = temp[col][(temp[col]<=LowerBound)|(temp[col]>=UpperBound)]

OutlierFreeData = data.drop(outlier.index,axis=0)
print("\nno. of Outlier",len(outlier), "\n% of Outlier = ", str(len(outlier)/
 ↪len(data)*100))
```

```
Before Missing Data Treatment
Glucose  =  5
BloodPressure  =  35
BMI  =  11
DiabetesPedigreeFunction  =  0

After Missing Data Treatment
Glucose  =  0
BloodPressure  =  0
BMI  =  0
DiabetesPedigreeFunction  =  0

no. of Outlier 4
% of Outlier =  0.5208333333333333
```

```
C:\Users\nikhi\AppData\Local\Temp\ipykernel_1716\3315839205.py:8:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  data[col][data[col]==0] = data[col].median()
```

- Split the data into 80% training and 20% testing data. Then, use a Decision Tree classifier algorithm with target variable as 'Outcome'.
- Print the default model performance metrics: Accuracy, Precision, Recall, F1Score

[3]:
```
# Spliting of data 7:3
X = OutlierFreeData.drop("Outcome",axis=1)
```

```
y = OutlierFreeData["Outcome"]
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.20,␣
 ↪random_state=4)

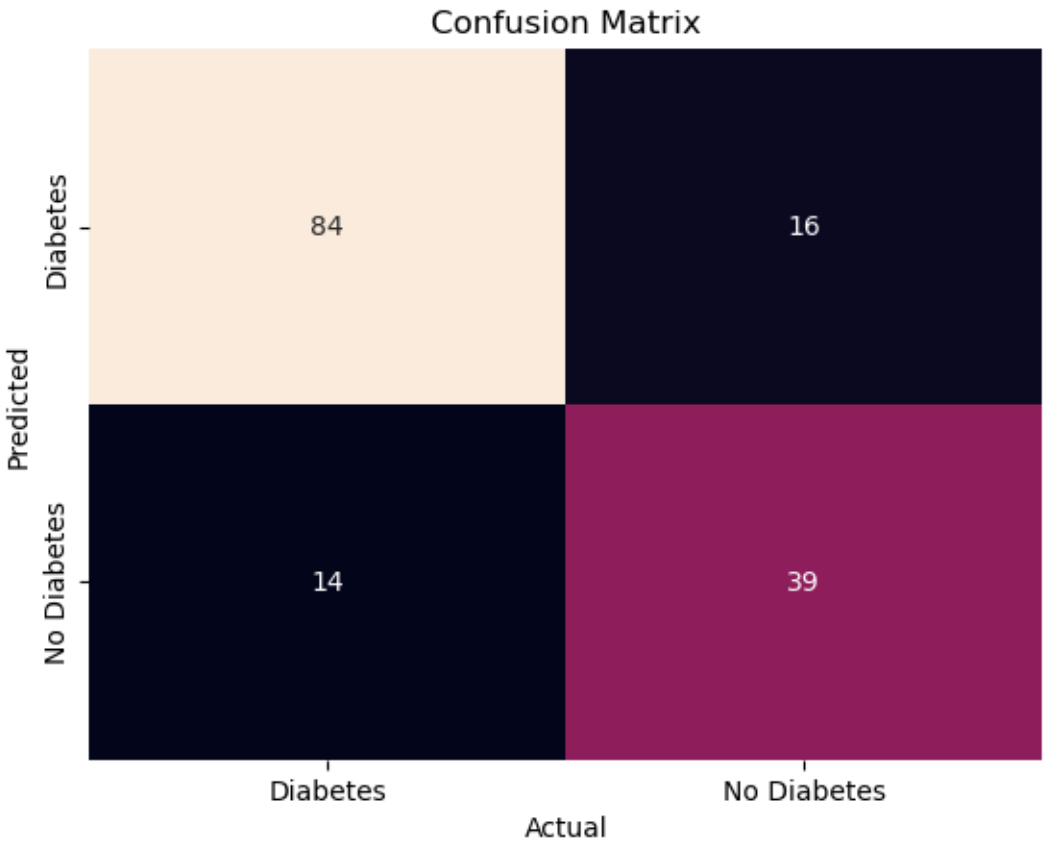
model = DecisionTreeClassifier(random_state = 64)
model.fit(X_train,y_train)
y_pred = model.predict(X_test)

print("Model Performance metrics are as below :-\n")
print("Accuracy is ",accuracy_score(y_test,y_pred))
print("Precision is ",precision_score(y_test,y_pred))
print("Recall is ",recall_score(y_test,y_pred))
print("F1-Score is ",f1_score(y_test,y_pred))

cm = confusion_matrix(y_test,y_pred)
print(cm)
ax = plt.subplot()
sns.heatmap(cm,annot = True , fmt = "d" , cbar = False , ax = ax)
ax.set_xlabel('Actual')
ax.set_ylabel('Predicted')
ax.set_title('Confusion Matrix')
ax.xaxis.set_ticklabels(['Diabetes','No Diabetes'])
ax.yaxis.set_ticklabels(['Diabetes','No Diabetes'])
plt.show()
print(tree.export_text(model))
```

```
Model Performance metrics are as below :-

Accuracy is  0.803921568627451
Precision is  0.7090909090909091
Recall is  0.7358490566037735
F1-Score is  0.7222222222222221
[[84 16]
 [14 39]]
```




```
|--- feature_1 <= 127.50
|   |--- feature_5 <= 28.50
|   |   |--- feature_3 <= 30.95
|   |   |   |--- feature_4 <= 0.67
|   |   |   |   |--- class: 0
|   |   |   |--- feature_4 >  0.67
|   |   |   |   |--- feature_3 <= 24.05
|   |   |   |   |   |--- class: 1
|   |   |   |   |--- feature_3 >  24.05
|   |   |   |   |   |--- class: 0
|   |   |--- feature_3 >  30.95
|   |   |   |--- feature_4 <= 0.90
|   |   |   |   |--- feature_2 <= 94.00
|   |   |   |   |   |--- feature_3 <= 31.10
|   |   |   |   |   |   |--- feature_2 <= 65.00
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_2 >  65.00
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- feature_3 >  31.10
|   |   |   |   |   |   |--- feature_1 <= 89.50
```

```
|   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |--- feature_1 >  89.50
|   |   |   |   |   |   |   |   |--- feature_4 <= 0.14
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 118.00
|   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |--- feature_1 >  118.00
|   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_4 >  0.14
|   |   |   |   |   |   |   |   |   |--- feature_4 <= 0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 82.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 31.80
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  31.80
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  82.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_4 >  0.50
|   |   |   |   |   |   |   |   |   |   |--- feature_4 <= 0.51
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 41.75
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  41.75
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |--- feature_4 >  0.51
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 36.45
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  36.45
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 4
|   |   |   |   |   |--- feature_2 >  94.00
|   |   |   |   |   |   |--- feature_1 <= 119.00
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_1 >  119.00
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |--- feature_4 >  0.90
|   |   |   |   |   |--- feature_3 <= 32.65
|   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- feature_3 >  32.65
|   |   |   |   |   |   |--- feature_0 <= 2.50
|   |   |   |   |   |   |   |--- feature_1 <= 105.00
|   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |--- feature_1 >  105.00
|   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- feature_0 >  2.50
|   |   |   |   |   |   |--- class: 1
|   |--- feature_5 >  28.50
|   |   |--- feature_3 <= 26.35
|   |   |   |--- class: 0
|   |   |--- feature_3 >  26.35
|   |   |   |--- feature_1 <= 91.50
```

```
|   |   |   |   |--- feature_4 <= 1.16
|   |   |   |   |   |--- feature_5 <= 44.00
|   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- feature_5 >  44.00
|   |   |   |   |   |   |--- feature_3 <= 35.10
|   |   |   |   |   |   |   |--- feature_1 <= 72.50
|   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |--- feature_1 >  72.50
|   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_3 >  35.10
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |--- feature_4 >  1.16
|   |   |   |   |   |--- class: 1
|   |   |   |--- feature_1 >  91.50
|   |   |   |   |--- feature_4 <= 0.56
|   |   |   |   |   |--- feature_4 <= 0.20
|   |   |   |   |   |   |--- feature_3 <= 27.65
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_3 >  27.65
|   |   |   |   |   |   |   |--- feature_1 <= 97.00
|   |   |   |   |   |   |   |   |--- feature_4 <= 0.12
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |--- feature_4 >  0.12
|   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |--- feature_1 >  97.00
|   |   |   |   |   |   |   |   |--- feature_5 <= 52.50
|   |   |   |   |   |   |   |   |   |--- feature_1 <= 102.50
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 79.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  79.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_1 >  102.50
|   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |--- feature_5 >  52.50
|   |   |   |   |   |   |   |   |   |--- feature_5 <= 58.50
|   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_5 >  58.50
|   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- feature_4 >  0.20
|   |   |   |   |   |   |--- feature_5 <= 54.50
|   |   |   |   |   |   |   |--- feature_4 <= 0.38
|   |   |   |   |   |   |   |   |--- feature_3 <= 34.05
|   |   |   |   |   |   |   |   |   |--- feature_2 <= 87.00
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 9.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_0 >  9.50
|   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |--- feature_2 >  87.00
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 95.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  95.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_3 >  34.05
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_5 <= 30.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |   |--- feature_5 >  30.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  1.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 40.35
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 3
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  40.35
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_4 >  0.38
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 2.00
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 68.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  68.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_0 >  2.00
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 12.50
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |   |--- feature_0 >  6.50
|   |   |   |   |   |   |   |   |   |   |   |   |--- truncated branch of depth 2
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  12.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |--- feature_5 >  54.50
|   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- feature_4 >  0.56
|   |   |   |   |   |   |--- feature_0 <= 0.50
|   |   |   |   |   |   |   |--- feature_5 <= 53.00
|   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |--- feature_5 >  53.00
|   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_0 >  0.50
|   |   |   |   |   |   |   |--- feature_5 <= 42.00
|   |   |   |   |   |   |   |   |--- feature_5 <= 33.50
|   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_5 >  33.50
|   |   |   |   |   |   |   |   |   |--- feature_4 <= 0.73
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 4.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  4.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_4 >  0.73
```

```
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |--- feature_5 >  42.00
|   |   |   |   |   |   |   |--- class: 1
|--- feature_1 >  127.50
|   |--- feature_3 <= 29.95
|   |   |--- feature_1 <= 151.50
|   |   |   |--- feature_0 <= 1.50
|   |   |   |   |--- class: 0
|   |   |   |--- feature_0 >  1.50
|   |   |   |   |--- feature_2 <= 73.00
|   |   |   |   |   |--- feature_0 <= 4.50
|   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- feature_0 >  4.50
|   |   |   |   |   |   |--- class: 0
|   |   |   |   |--- feature_2 >  73.00
|   |   |   |   |   |--- feature_4 <= 0.18
|   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- feature_4 >  0.18
|   |   |   |   |   |   |--- class: 0
|   |   |--- feature_1 >  151.50
|   |   |   |--- feature_5 <= 59.00
|   |   |   |   |--- feature_5 <= 25.50
|   |   |   |   |   |--- class: 0
|   |   |   |   |--- feature_5 >  25.50
|   |   |   |   |   |--- feature_3 <= 27.10
|   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- feature_3 >  27.10
|   |   |   |   |   |   |--- feature_5 <= 36.50
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_5 >  36.50
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |--- feature_5 >  59.00
|   |   |   |   |--- class: 0
|   |--- feature_3 >  29.95
|   |   |--- feature_1 <= 157.50
|   |   |   |--- feature_5 <= 30.50
|   |   |   |   |--- feature_2 <= 73.00
|   |   |   |   |   |--- feature_3 <= 32.45
|   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- feature_3 >  32.45
|   |   |   |   |   |   |--- feature_0 <= 0.50
|   |   |   |   |   |   |   |--- feature_3 <= 33.75
|   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |--- feature_3 >  33.75
|   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_0 >  0.50
|   |   |   |   |   |   |   |--- feature_1 <= 136.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 131.50
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_2 <= 56.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |   |--- feature_2 >  56.00
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |--- feature_1 >  131.50
|   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_1 >  136.50
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |--- feature_2 >  73.00
|   |   |   |   |   |--- feature_3 <= 41.80
|   |   |   |   |   |   |--- feature_0 <= 4.50
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |--- feature_0 >  4.50
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- feature_3 >  41.80
|   |   |   |   |   |   |--- feature_4 <= 0.86
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_4 >  0.86
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |--- feature_5 >  30.50
|   |   |   |   |--- feature_4 <= 0.43
|   |   |   |   |   |--- feature_3 <= 32.45
|   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- feature_3 >  32.45
|   |   |   |   |   |   |--- feature_1 <= 130.50
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |--- feature_1 >  130.50
|   |   |   |   |   |   |   |--- feature_0 <= 9.50
|   |   |   |   |   |   |   |   |--- feature_2 <= 79.00
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 6.50
|   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 46.65
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |--- feature_3 >  46.65
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_0 >  6.50
|   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_2 >  79.00
|   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |--- feature_0 >  9.50
|   |   |   |   |   |   |   |   |--- feature_1 <= 141.50
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |--- feature_1 >  141.50
|   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |--- feature_4 >  0.43
|   |   |   |   |   |--- feature_5 <= 68.00
|   |   |   |   |   |   |--- feature_1 <= 152.50
|   |   |   |   |   |   |   |--- feature_4 <= 1.55
|   |   |   |   |   |   |   |   |--- feature_2 <= 67.00
```

```
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 8.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  8.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_2 >  67.00
|   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 40.05
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |   |--- feature_3 >  40.05
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 <= 41.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |   |--- feature_3 >  41.00
|   |   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_4 >  1.55
|   |   |   |   |   |   |   |   |   |--- feature_3 <= 41.80
|   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |--- feature_3 >  41.80
|   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |--- feature_1 >  152.50
|   |   |   |   |   |   |   |   |--- feature_3 <= 36.50
|   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_3 >  36.50
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |--- feature_5 >  68.00
|   |   |   |   |   |   |   |--- class: 0
|   |   |   |--- feature_1 >  157.50
|   |   |   |   |--- feature_3 <= 46.10
|   |   |   |   |   |--- feature_4 <= 2.23
|   |   |   |   |   |   |--- feature_4 <= 0.30
|   |   |   |   |   |   |   |--- feature_4 <= 0.27
|   |   |   |   |   |   |   |   |--- feature_5 <= 56.50
|   |   |   |   |   |   |   |   |   |--- feature_0 <= 9.50
|   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |   |--- feature_0 >  9.50
|   |   |   |   |   |   |   |   |   |   |--- feature_0 <= 11.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |   |   |--- feature_0 >  11.50
|   |   |   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |   |--- feature_5 >  56.50
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |--- feature_4 >  0.27
|   |   |   |   |   |   |   |   |--- feature_5 <= 37.50
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |--- feature_5 >  37.50
|   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |--- feature_4 >  0.30
|   |   |   |   |   |   |   |--- feature_5 <= 43.00
|   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |--- feature_5 >  43.00
```

```
|   |   |   |   |   |   |   |--- feature_5 <= 50.50
|   |   |   |   |   |   |   |   |--- feature_0 <= 7.50
|   |   |   |   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |   |   |--- feature_0 >  7.50
|   |   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |   |   |--- feature_5 >  50.50
|   |   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- feature_4 >  2.23
|   |   |   |   |   |   |--- class: 0
|   |   |   |--- feature_3 >  46.10
|   |   |   |   |--- feature_1 <= 163.50
|   |   |   |   |   |--- class: 1
|   |   |   |   |--- feature_1 >  163.50
|   |   |   |   |   |--- feature_1 <= 176.50
|   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- feature_1 >  176.50
|   |   |   |   |   |   |--- class: 1
```

- Plot a Precision & Recall vs max_leaf_nodes (consider a range of numbers) curve (both Prec and Rec on the same graph). Find the kernel type for which F1-score is the highest.
- Plot a Precision & Recall vs max_depth (consider a range of numbers) curve (both Prec and Rec on the same graph). Find the kernel type for which F1-score is the highest.

[4]:
```
criterion = ['gini','entropy', 'log_loss']

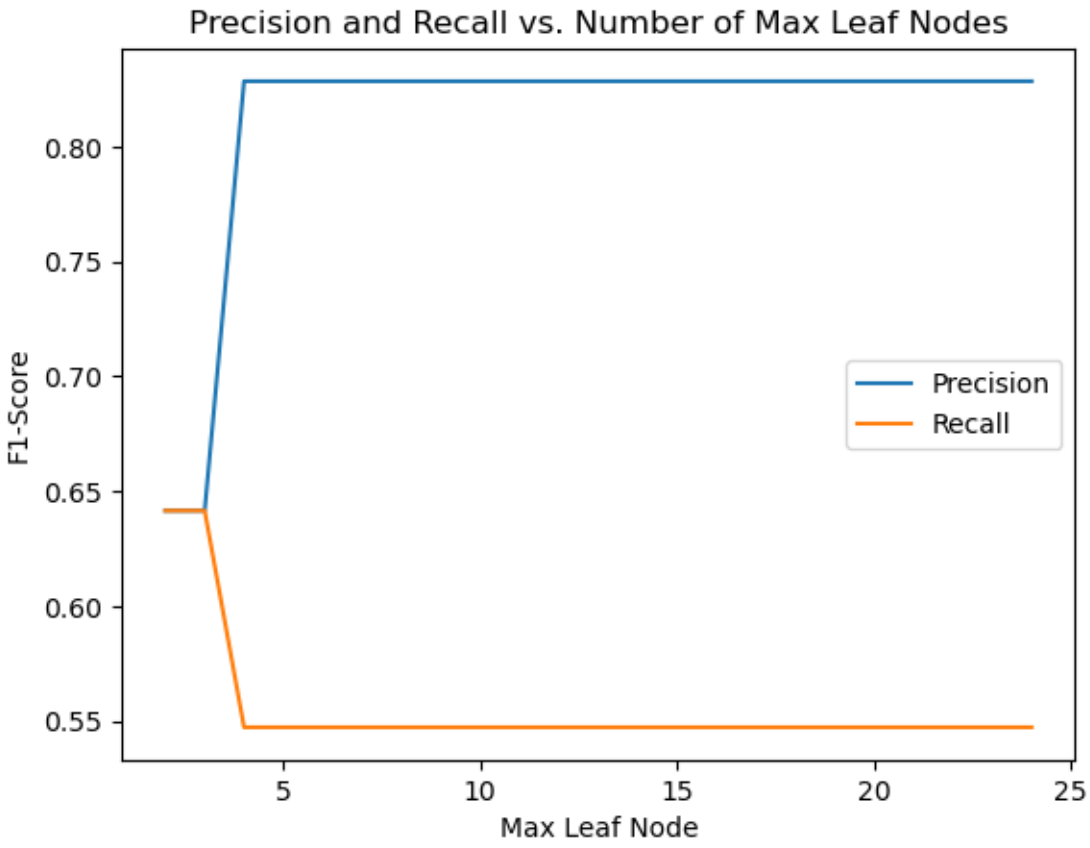
df = pd.DataFrame(columns=['Criterion','Max Tree Depth','Max Leaf␣
 ↪Nodes','precision','recall','f1-Score'])
for c in criterion:
    for md in range(1,25):
        for mln in range(2,25):
                model = DecisionTreeClassifier(criterion = c, max_depth = md,␣
 ↪max_leaf_nodes = mln, random_state = 64)
                model = model.fit(X_train,y_train)
                y_pred = model.predict(X_test)
                df = pd.concat([pd.DataFrame([[c, md, mln,␣
 ↪precision_score(y_test,y_pred), recall_score(y_test,y_pred),␣
 ↪f1_score(y_test,y_pred)]], columns=df.columns), df], ignore_index=True)

display(df[(df['f1-Score'] == max(df['f1-Score']))])
```

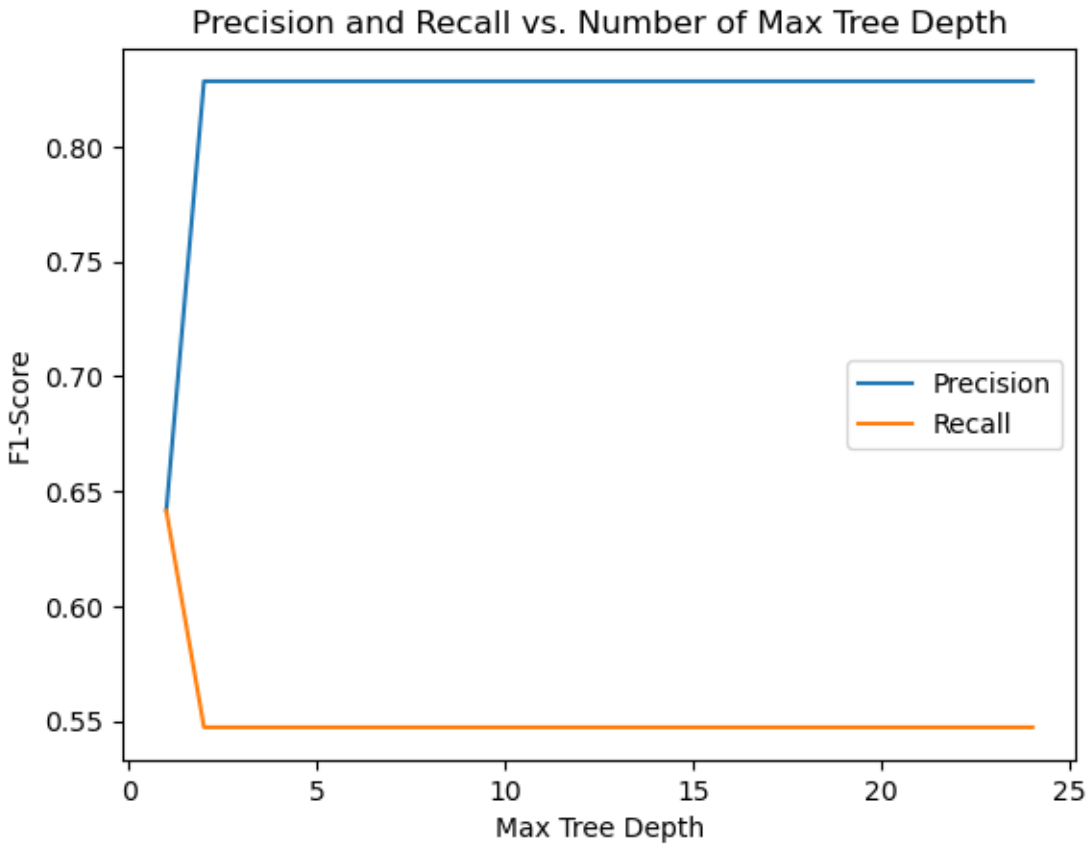
	Criterion	Max Tree Depth	Max Leaf Nodes	precision	recall	f1-Score
1107	gini	24	21	0.759259	0.773585	0.766355
1108	gini	24	20	0.759259	0.773585	0.766355
1130	gini	23	21	0.759259	0.773585	0.766355
1131	gini	23	20	0.759259	0.773585	0.766355
1153	gini	22	21	0.759259	0.773585	0.766355
1154	gini	22	20	0.759259	0.773585	0.766355

1176	gini	21	21	0.759259	0.773585	0.766355
1177	gini	21	20	0.759259	0.773585	0.766355
1199	gini	20	21	0.759259	0.773585	0.766355
1200	gini	20	20	0.759259	0.773585	0.766355
1222	gini	19	21	0.759259	0.773585	0.766355
1223	gini	19	20	0.759259	0.773585	0.766355
1245	gini	18	21	0.759259	0.773585	0.766355
1246	gini	18	20	0.759259	0.773585	0.766355
1268	gini	17	21	0.759259	0.773585	0.766355
1269	gini	17	20	0.759259	0.773585	0.766355
1291	gini	16	21	0.759259	0.773585	0.766355
1292	gini	16	20	0.759259	0.773585	0.766355
1314	gini	15	21	0.759259	0.773585	0.766355
1315	gini	15	20	0.759259	0.773585	0.766355
1337	gini	14	21	0.759259	0.773585	0.766355
1338	gini	14	20	0.759259	0.773585	0.766355
1360	gini	13	21	0.759259	0.773585	0.766355
1361	gini	13	20	0.759259	0.773585	0.766355
1383	gini	12	21	0.759259	0.773585	0.766355
1384	gini	12	20	0.759259	0.773585	0.766355
1406	gini	11	21	0.759259	0.773585	0.766355
1407	gini	11	20	0.759259	0.773585	0.766355
1429	gini	10	21	0.759259	0.773585	0.766355
1430	gini	10	20	0.759259	0.773585	0.766355
1453	gini	9	20	0.759259	0.773585	0.766355

[10]:
```
# Plotting precision and recall scores against Max Leaf Nodes
y = df[df['Max Tree Depth']==2]
x = y[y['Criterion']=='gini']
plt.plot(x["Max Leaf Nodes"], x["precision"], label='Precision')
plt.plot(x["Max Leaf Nodes"], x["recall"], label='Recall')
plt.xlabel('Max Leaf Node')
plt.ylabel('F1-Score')
plt.title('Precision and Recall vs. Number of Max Leaf Nodes')
plt.legend()
plt.show()
```



[12]:
```
# Plotting precision and recall scores against Max Tree Depth
y = df[df['Max Leaf Nodes']==4]
x = y[y['Criterion']=='gini']
plt.plot(x["Max Tree Depth"], x["precision"], label='Precision')
plt.plot(x["Max Tree Depth"], x["recall"], label='Recall')
plt.xlabel('Max Tree Depth')
plt.ylabel('F1-Score')
plt.title('Precision and Recall vs. Number of Max Tree Depth')
plt.legend()
plt.show()
```



[7]:
```
model = DecisionTreeClassifier(criterion = 'gini', max_depth = 9,␣
 ↪max_leaf_nodes = 20, random_state = 64)
model = model.fit(X_train,y_train)
y_pred = model.predict(X_test)
print("Model Performance metrics are as below :-")
print("Accuracy is ",accuracy_score(y_test,y_pred))
print("Precision is ",precision_score(y_test,y_pred))
print("Recall is ",recall_score(y_test,y_pred))
print("F1-Score is ",f1_score(y_test,y_pred))
```

```
Model Performance metrics are as below :-
Accuracy is  0.8366013071895425
Precision is  0.7592592592592593
Recall is  0.7735849056603774
F1-Score is  0.7663551401869159
```

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